

Supplementary Information of

CoupledMD: molecular dynamics trajectories of active GPCR–G-protein complexes

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Supplementary Note 1. Cohort definition and release controls

This note documents the release controls behind the cohort summarized in the main text. Gs_8HTI is a consensus OR52c model with no canonical UniProt entry and is retained with explicit null receptor-accession and gene fields; it is the reason the release reports 174 distinct receptor names but only 173 mapped UniProt accessions.

The thirteen unresolved records break down by category as periodic-boundary/trajectory-continuity (8), component-continuity (4) and incomplete-trajectory evidence (1); Gq_7E9W is the single record excluded outright, as a non-GPCR duplicate/mislabel of Gq_8E9W. Timestamp resets are treated as diagnostic only and are not classified as failures. Record-level evidence is provided in Supplementary Data S2.

The following release assertions are verified before the archive is frozen:

Assertion	Validated result
Unique final system identifiers	207
Selected production replicas	621
Total selected production sampling	310,500 ns
Final systems covered by each current portal/API record type	207
Unresolved/excluded IDs in final derived tables	0
Gq_7E9W in final cohort/API records	0
Gq_7DWC in final cohort/API records	0
Pocket records	2,149 detected rows + 2 valid zero-pocket rows
Gateway records	5,796 system/interface/metric rows
Frozen archive, DOI and archive checksums	Pending final manifest freeze and deposition
Package schema and integrity assertions	Passed

Supplementary Note 2. Structure provenance and protocol groups

Supplementary Data S1 serializes the final 207-system identifier, source PDB identifier, receptor name and mapped accession/gene when available, GPCR class, G-protein family and subtype label, authoritative ligand name and chemical identifier when present, selected-replica counts and durations, force-field/protocol label, lipid composition, structural-provenance label, bilayer/trajectory flags and source-availability flags. It does not serialize activation state, biological assembly, retained-chain lists or retrieval dates because those fields are not supported consistently by the authoritative sources.

The release defines three protocol groups:

Protocol	Systems	Engine	Current label
P1	180	AMBER pmemd	CHARMM36 (chamber, AMBER pmemd)
P2	26	AMBER (CHARMM-GUI)	CHARMM36 (via AMBER CHARMM-GUI)
P3	2	GROMACS	CHARMM36 (membrane-embedded, GROMACS)

The shared thermodynamic and non-bonded settings, and the engine-specific thermostat and barostat algorithms, are given in the main Methods; only the per-group force-field labeling above is release-specific.

Supplementary Data S10 inventories the current plotting, package-build and API code, repository-relative paths, byte sizes, SHA-256 values, languages and versions available in the package-build environment. No authoritative record of the exact historical AMBER or GROMACS engine versions was located; therefore the supplement does not claim exact simulation-engine versions.

Supplementary Note 3. Production and reduced trajectory records

Supplementary Data S4 is a 414-row source-inventory evidence table with one aggregate production-trajectory source row and one topology source row per final system. It reports limited source locators, existence and byte size observed during audit, and the three-replica/500-ns release scope. S4 is not a frozen archive manifest and does not claim one row per deposited physical file, repository paths, per-file checksums or DOI deposition.

Supplementary Data S7 is the 621-row selected-replica release ledger and records three 500-ns selected replicas per system. The complete 621-file trajectory manifest, matched per-file topology identifiers, checksums and a self-contained file-level QC package remain pending final manifest freeze and deposition.

Supplementary Note 4. Pocket workflow

Ligand heavy atoms were identified from retained HETATM records after excluding protein, lipid, water and ion residue names. The per-cluster metric `mean_freq` is the mean occupied-frame fraction of its grid points; `best_ortho_freq` (defined in the main Methods) is the maximum `mean_freq` among clusters labelled orthosteric.

Supplementary Data S5 contains 2,149 detected-pocket rows from 205 systems. Gi_7YK6 and Gi_8YIC have valid API pocket records with `n_pockets=0` and are represented by two explicit `available_zero_pockets` rows rather than by numerical-zero pocket measurements or unavailable records. Detected rows report pocket size, mean frequency, orthosteric annotation, stored location, reader-facing anatomical zone and mapped GPCRdb generic positions.

Supplementary Note 5. Gateway workflow

The seven adjacent helix pairs scored for lipid access are TM1–TM2, TM2–TM3, TM3–TM4, TM4–TM5, TM5–TM6, TM6–TM7 and TM7–TM1, with TM boundaries taken from GPCRdb assignments.

Supplementary Data S6 contains 5,796 tidy system-interface-metric rows (207 systems x seven adjacent TM-helix pairs x four metrics). The metrics are penetration and `penetration_p90` (angstrom), `open_fraction` (unitless fraction) and occupancy (lipid heavy atoms per frame). Every row reports the three replica values, their mean, stored interval bounds, replica count, unit and schema version. The 95% intervals were obtained by resampling the three replica summaries 1,000 times under a fixed seed.

Supplementary Note 6. Technical validation

The production-trajectory acceptance criteria, orthosteric-pocket-recovery validation and reduced-visualization audit are described in the main Technical Validation. Supplementary Data S7 is the 621-row selected-replica ledger. Supplementary Data S8 is a 207-system readiness/QC summary with pointers to the underlying evidence; it is not a self-contained full QC package. Supplementary Data S9 summarizes current

portal/API file and content availability. A valid zero-pocket JSON record counts as available, and API file presence is not interpreted as deposited-archive coverage.

Supplementary Table S1. Canonical metadata dictionary

Supplementary Data S3 contains one row for each of the 105 unique field names actually serialized across S1-S10. Every row reports field, entity_level, type, unit, requiredness, nullable, allowed_vocabulary, definition, example, provenance_authority and schema_version. Hypothetical archive-manifest fields are not listed.

S3 field	Type	Requiredness	Meaning
field	string	required	Exact column name serialized in S1-S10
entity_level	string	required	Actual record level at which the field is interpreted
type	string	required	Logical serialized type
unit	string	required	Measurement unit or explicit unitless/not-applicable label
requiredness	enum	required	required or conditional
nullable	boolean	required	Whether an empty CSV value is permitted
allowed_vocabulary	string	conditional	Pipe-delimited closed vocabulary when applicable
definition	string	required	Reader-facing semantic definition
example	string	conditional	Example non-null serialized value
provenance_authority	string	required	Authoritative source or computation
schema_version	string	required	Schema identifier v9-final207.1

Empty strings, literal nan, zero and absent keys are not interchangeable; each serialization defines the null representation.

Supplementary Table S2. Unresolved and excluded records

Supplementary Data S2 combines the 13 unresolved records and the single excluded record, Gq_7E9W, in one 14-row release-boundary table. There is no separate exclusions file. Neither unresolved nor excluded records contribute to final-cohort derived tables.

Supplementary Table S3. Molecular-file specification

Record role	Format	Required pairing/source	Current interpretation
Production-trajectory source evidence	NetCDF or TRR	Matched source topology	Aggregate S4 evidence; not a deposited per-file manifest

Record role	Format	Required pairing/source	Current interpretation
Topology source evidence	PRMTOP/PARM7 or TPR	Production trajectory	Aggregate S4 evidence; not a deposited per-file manifest
Reduced reference	PDB	Reduced XTC	Current portal visualization
Reduced trajectory	XTC	Reduced PDB	Visualization and lightweight inspection
Pocket/gateway records	JSON/NPZ and CSV	Per-system schema version	Current derived annotations
Readiness/QC summary	CSV	Evidence pointers	Release evidence summary, not full per-file QC

Supplementary Table S3 distinguishes the molecular-record roles relevant to the current server and future deposition. The current Supplementary Data package contains aggregate source evidence and derived CSV records; frozen per-file format metadata, pairings, checksums and recommended-reader details remain pending archive-manifest freeze.

Supplementary Table S4. API and versioning contract

The current OpenAPI schema describes the final-207 API access layer. Supplementary Data S9 reports ten current portal/API record types using explicit file/content availability semantics, including valid zero-pocket JSON records. These counts describe the current server and do not demonstrate a deposited archive or frozen file manifest.

Supplementary Figure S1 | Receptor representation across G-protein families. Circular dendrogram of 174 receptor names clustered by Jaccard distance and average linkage using binary family-presence profiles. Leaves are labeled by representative PDB identifiers and colored by G-protein family; gray denotes receptors represented with multiple families. The asterisk identifies Gs_8HTI, which lacks a mapped UniProt accession. Branch proximity reflects release coverage only, not sequence or structural similarity.

Supplementary Figure S2 | Definition and aggregation of transmembrane-gateway records. (A) Seven gateways formed by adjacent TM-helix pairs and the associated record-identity requirements. (B) A lipid heavy atom is classified as wedged when it lies within 5.0 Å of both helices and inside the central TM z-band; a frame is open when the maximum inward penetration, p_{max} , is at least 0.5 Å. (C) Frame-level measurements are summarized per replica and combined across three replicas to produce penetration, penetration-P90, open-fraction and occupancy records with stored 95% intervals.

Supplementary Data files

File	Content	Granularity
S1	Included-system inventory; ligand fields populated only from authoritative metadata	207 systems
S2	Combined release-boundary table: 13 unresolved and two excluded records	15 records
S3	Dictionary for every field actually serialized in S1-S10	105 unique fields
S4	Production/topology source-inventory evidence; not an archive manifest	414 rows; two per final system
S5	Pocket summaries with GPCRdb mappings and explicit valid zero-pocket records	2,149 pocket rows + 2 zero-pocket rows
S6	Tidy gateway means, intervals and three replica values for four metrics	5,796 system/interface/metric rows
S7	Selected-replica release ledger; not frozen file-level QC	621 system/replica rows
S8	Final-cohort readiness/QC summary with evidence pointers	207 systems
S9	Current portal/API file and content availability; not archive coverage	10 record types
S10	Current code files, environment files and package-build dependencies	24 inventory items

Supplementary reuse cautions

Split machine-learning datasets by receptor accession, not by replica, to avoid data leakage.

Treat `protocol_id`, GPCR class, ligand context, construct type and source PDB as potential confounders.

Do not infer physiological coupling prevalence from this structurally determined cohort.

Do not use reduced records for lipid, ion, solvent or fast-kinetics analyses.

occupancy is a count of wedged lipid atoms, not a distance; do not label it as Å.

Cite the Scientific Data article and identify the CoupledMD server access date. Once a DOI-linked immutable archive is deposited, also cite its exact DOI and version.

Supplementary Data Availability

Current final-207 records are available at <https://www.coupleddmd.cn> (API documentation: `/api/docs`). The frozen manifest, checksums and DOI-linked archive remain pending deposition; no archive-completeness, licensing or direct-download claim is made.